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Common Mode Voltage Reduction of Single-Phase Quasi-Z-Source Inverter-Based Photovoltaic System

YUSHAN LIU¹, (Senior Member, IEEE), BAOMING GE², (Member, IEEE),
XIAO LI², (Member, IEEE), AND YAOSUO XUE³, (Senior Member, IEEE)

¹School of Automation Science and Electrical Engineering, Beihang University, Beijing 100083, China

²Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX 77843, USA

³Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA

Corresponding author: Xiao Li (xiaoli.tamu@gmail.com)

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ABSTRACT A common mode voltage (CMV) reduction method is proposed for the active power filter (APF) integrated single-phase quasi-Z-source inverter (qZSI) based photovoltaic (PV) power system. The APF integrated qZSI completely compensates the dc-side double-line-frequency (2ω) ripple through the APF capacitor, while maintaining low qZS inductance and capacitance. The proposed method reduces the CMV amplitude to half of that using the traditional sinusoidal pulse width modulation (SPWM). The switching and conduction power losses are analyzed. Simulation and experimental results demonstrate the proposed method reduces CMV without comprising performance and efficiency of the APF integrated single-phase qZSI.

INDEX TERMS Common mode voltage, single-phase inverter, quasi-Z-source inverter, pulsewidth modulation.

I. INTRODUCTION

Quasi-Z-source inverter (qZSI) is attracting high interests in the application of Photovoltaic (PV) power systems. The wide operating range of dc input voltage makes it capable of dealing with the voltage variations of PV panels, and connecting to the power grid without a bulky step-up transformer, while maintaining the single-stage inverter topology with simple structure and control design [1]–[5]. Furthermore, qZSI's shoot-through behavior eliminates the deadtime concern and improves the anti-electromagnetic interference capability [6].

The active power filter (APF) integrated single-phase qZSI is expected to be a powerful backup for the residential PV power systems. A half-bridge phase leg and a compensation capacitor formed APF branch fully decouples the dc-side double-line-frequency (2ω) ripple [4]. As a result, the ac output quality is improved; the APF capacitance and qZS inductance and capacitance are small in value and volume [7]–[13], because the APF branch allows highly fluctuated ac voltage and current, and the qZS inductance and capacitance

are only utilized to filter the switching frequency voltage and current ripples. By using Silicon Carbide (SiC) power devices, it is possible to further reduce the passive component values through increasing the switching frequency [12], [13].

Whereas, due to lack of galvanic isolation in such transformer-less PV power systems, the coupling among the PV panel stray capacitance, inverter, output filter, and grid impedance causes a common mode resonant path [14]–[16]. Hence, there will be leakage current flowing into the ground through this path because of a common mode voltage (CMV) with high du/dt rate and high amplitude [17]. For safety purpose, the leakage current should be suppressed within a certain value to guarantee safety of person. This is also proved to be beneficial to reduce the possible extra power losses, slower the degradation of PV panels, and improve grid current quality [18]–[21]. The leakage current values can be impacted by modulation methods, CMV conditions, and stray capacitances, etc. The stray capacitance is affected by several factors, such as PV panel and chassis structure, the PV cell surface, the distance between cells, weather conditions, and type of common-mode filter [15], [16]. Contributions have been conducted on development of new topologies or

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solutions to lower the leakage current through modifying the CMV amplitude [20]. Especially, when utilizing SiC power devices with high switching frequency, it becomes critical to suppress leakage current by low or constant CMV amplitudes, which helps to mitigate the system high-frequency emission source [22].

For traditional single-phase inverter, the CMV reduction technology can be categorized as modulation methods modification and topology improvement. The former reconstructs the voltage vectors to reduce the leakage current amplitude which is well-known by alternating the switching combinations under space vector modulation (SVM) [23]–[25], the latter commonly adds extra power devices to decouple the leakage current path [26]. Similarly, in order to reduce CMV of the qZSI, the SVM was revised [27]–[29], and a reverse diode was added in the return path of the circuit [30], but they are coordinated for the three-phase inverter topology. For single-phase qZSI, the sinusoidal pulse width modulation (SPWM) is commonly used and the switching alternation to reduce CMV has not been investigated yet. Semiconductor devices were added in the ac output of the single-phase qZSI to reduce the CMV [31], but extra cost and complex circuit are inevitable.

This paper proposes a modified modulation method for the APF integrated single-phase qZSI PV power system, to reduce the CMV amplitude while suppressing the leakage current without extra devices. The organization of the paper is as follows: the APF integrated single-phase qZSI PV system is introduced and the common-mode model is established in Section II; the proposed CMV reduction method is detailed in Section III; Section IV demonstrates simulation results; finally, Section V concludes the work.

II. APF INTEGRATED SINGLE-PHASE QZSI PV SYSTEM

Fig. 1 shows the APF integrated single-phase qZSI-based PV power system [7]. Two inductors L_1 and L_2 , two capacitors C_1 and C_2 , and one diode D_1 constructed qZS network is inserted into the dc link of a conventional single-phase inverter. The bridge legs 1 and 2 represent the load power phase legs, connecting to the single-phase ac load or grid through a filter inductor L_f . The half-bridge leg 3,

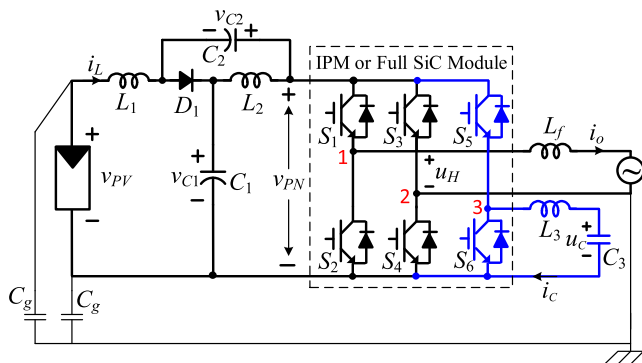


FIGURE 1. APF integrated single-phase qZSI PV power system.

a compensating capacitor C_3 , and a filter inductor L_3 form the APF branch. Hence, the 2ω ripple power is eliminated through the proper operation of switches S_5 and S_6 [7]–[12]. In addition, the C_g represents the stray capacitance between the PV panel and the ground.

A. OPERATING PRINCIPLE

The load power phase legs work the same as the conventional qZSI. The dc-link peak voltage V_{PN} , average input current I_L , and average qZS capacitor voltages V_{C1} and V_{C2} are, respectively

$$\begin{aligned} V_{PN} &= \frac{1}{1-2D} v_{PV}, \quad I_L = \frac{P_o}{v_{PV}} \\ V_{C1} &= \frac{1-D}{1-2D} v_{PV}, \quad V_{C2} = \frac{D}{1-2D} v_{PV} \end{aligned} \quad (1)$$

where v_{PV} denotes the PV panel voltage, D denotes the ST duty cycle, and P_o denotes the output power of qZSI.

The APF power devices S_5 and S_6 operate independently from the load power phase legs 1 and 2, where there is a phase shift between qZSI output voltage and APF capacitor voltage depending on the load conditions [32]. To completely buffer the 2ω ripple power, the APF branch's power P_{LC} should match

$$P_{LC}(t) = u_C(t)i_C(t) + L_3 i_C(t) \frac{di_C(t)}{dt} = -P_{2\omega}(t) \quad (2)$$

Therefore, the APF capacitor voltage can be derived as [13]

$$u_C(t) = \sqrt{\frac{P_m + P_{dc} \sin(2\omega t - \varphi)}{\omega C_3}} \quad (3)$$

where u_C and i_C denote the voltage and current of the compensating branch capacitor C_3 , $P_{2\omega}$ denotes the 2ω ripple power, P_m denotes the rated power of qZSI, $P_{dc} = P_o - P_{2\omega}$ denotes the dc component of qZSI's output power P_o , ω denotes the fundamental angular frequency of ac output, and φ denotes the power factor angle. It can be seen that the APF capacitor voltage is dc with a 2ω component.

B. COMMON-MODE MODEL

From Fig. 1, the common-mode model of the single-phase qZSI PV system at the switching-frequency range is obtained in Fig. 2. Fig. 2 (a) shows the equivalent model and Fig. 2 (b) shows the simplified model. The R_g represents the resistance between the PV panel and the ground, R_f represents the internal resistance of filter inductance L_f , and Z_g represents the ground impedance mostly resistive at several Ohms [28]. The CMV and the leakage current are represented by U_{CM} and i_{LK} , respectively.

The CMV U_{CM} of the single-phase qZSI is related to the output voltages of load power phase legs 1 and 2, in the form of [24]

$$U_{CM} = \frac{U_{10} + U_{20}}{2} \quad (4)$$

where U_{10} and U_{20} denote the voltage of phase legs 1 and 2 with respect to the ground, respectively.

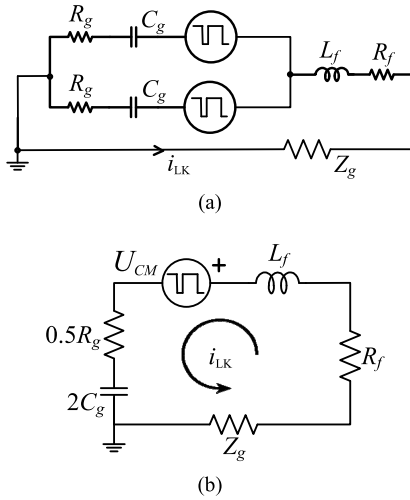


FIGURE 2. Common-mode model of the single-phase qZSI PV power system. (a) Equivalent model; (b) simplified model.

Due to the CMV, the leakage current flows into the ground, as shown in Fig. 2. Therefore, the PV panel leakage current in frequency domain can be obtained as

$$i_{lk}(\omega_s) = \frac{U_{cm}(\omega_s)}{0.5R_g + R_f + j\omega_s L_f + \frac{1}{2j\omega_s C_g}} \quad (5)$$

where $\omega_s = 2\pi f_s$ related to the switching frequency f_s .

And the transfer function from the leakage current to the CMV can be derived as

$$\frac{i_{lk}(s)}{U_{cm}(s)} = \frac{2C_g s}{2C_g L_f s^2 + (R_g + 2R_f) C_g s + 1} \quad (6)$$

demonstrating that reducing or maintaining constant common mode voltage amplitude is able to reduce the leakage current value.

III. PROPOSED METHOD OF COMMON MODE VOLTAGE REDUCTION

From the established common-mode leakage current model of (5) and (6), constant CMV or low CMV amplitude are options to reduce the leakage current while enhance the safety of the qZSI PV system. Whereas, it is hard to make CMV constant if without extra power devices for the single-phase inverter topology. Here, the proposed method uses a modified modulation method to reduce the CMV value, and the switching and conduction power losses are computed, respectively, to investigate the proposed method reducing CMV while without loss of efficiency.

A. PROPOSED METHOD

The single-phase qZSI consists of the active state (ACT), zero state (ZS), and shoot-through state (ST). Fig. 3 shows the sketch map of switching operation with the CMV, under the traditional SPWM of qZSI, and Table 1 lists the summary.

As shown in Fig. 3 and Table 1, there are two ACT, two zero states, and one ST state. The ACT with switching state of [1 0 0 1] is denoted as ACT₁, that of [0 1 1 0] is denoted as

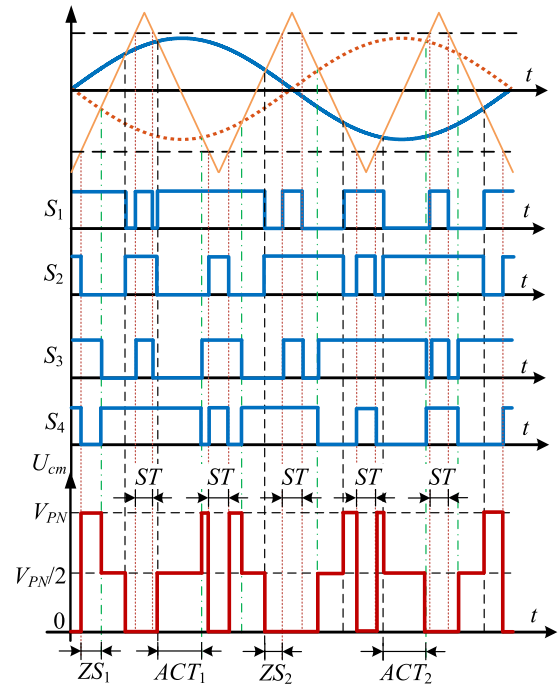


FIGURE 3. Sketch map of switching operation with the CMV of traditional SPWM of single-phase qZSI.

TABLE 1. Summary of operating states and CMV values of single-phase qZSI in the traditional SPWM.

States	S_1	S_2	S_3	S_4	CMV Values
ACT ₁	1	0	0	1	$V_{PN}/2$
ACT ₂	0	1	1	0	$V_{PN}/2$
ZS ₁	1	0	1	0	V_{PN}
ZS ₂	0	1	0	1	0
ST	1	1	1	1	0

ACT₂. Similarly, the ZS of [1 0 1 0] and [0 1 0 1] are denoted as ZS₁ and ZS₂, respectively. It can be seen that the CMV value is equal to zero at the ST and ZS₂, half of the dc-link peak voltage $V_{PN}/2$ at the ACT₁ and ACT₂, and the dc-link peak voltage V_{PN} at the ZS₁.

Therefore, we propose to revise all the ZS₁ into ZS₂, as the sketch map of Fig. 4 shows. Table 2 lists the operation of the MOSFET and diode of each power device under the proposed method. It can be seen that due to the eliminated ZS₁, in the positive half cycle of qZSI output current i_o , S_1 alternates in the ACT₁, ZS₂, and ST, the diode of S_2 freewheels in the ZS₂, S_4 is always ON, and the S_3 is OFF unless a ST state occurs. Similarly, in the negative half cycle of i_o , S_3 switches in the ACT₂, ZS₂, and ST, the diode of S_4 freewheels in the ZS₂, S_2 is ON, and S_1 switches at ST. In this way, the CMV amplitude is maintained at the half dc-link peak voltage $V_{PN}/2$, as shown in Fig. 4.

B. POWER LOSS CALCULATION

From Table 2 and Fig. 4, it can be obtained that in the positive half cycle of i_o , the MOSFET of S_1 switches between the

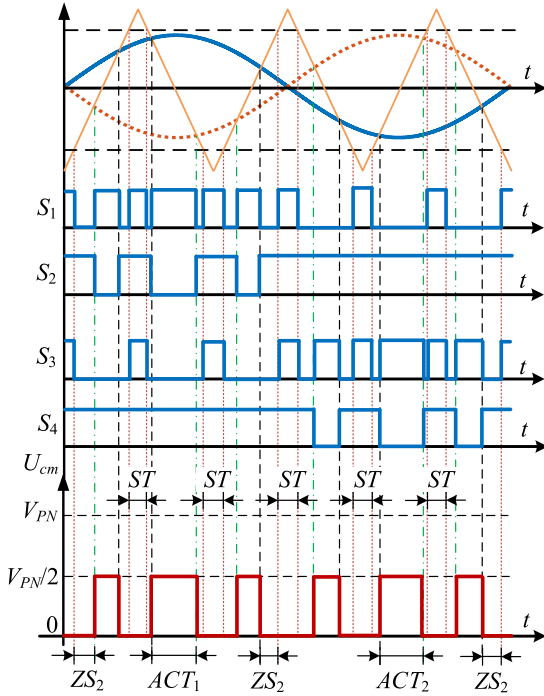


FIGURE 4. Sketch map of the proposed CMV reduction method for single-phase qZSI.

TABLE 2. Power devices operation in the proposed method.

		$i_o > 0$	$i_o < 0$
S_1	MOSFET diode	ACT ₁ , ZS ₂ , ST	ST switching
S_2	MOSFET diode	ACT ₁ , ZS ₂ , ST Freewheeling	ON
S_3	MOSFET diode	ST switching	ACT ₂ , ZS ₂ , ST
S_4	MOSFET diode	ON	ACT ₂ , ZS ₂ , ST Freewheeling

ACT and ZS, and the ST and ZS, the diode of S_2 changes between ACT and ZS, so that it freewheels in the ZS₂. There are two times ACT-ZS switching and two times ST-ZS switching of S_1 , and also two times ACT-ZS switching of S_2 , in each control cycle. In the negative half cycle of i_o , there are two times ST-ZS switching per control cycle while no ACT-ZS actions of S_1 ; the S_2 has no any switching behavior. Similar results can be obtained for S_3 and S_4 . Table 3 lists the comparison of switching times of the proposed method and traditional SPWM. It can be seen that in the traditional SPWM, all devices operate equally for one time ACT-ZS switching and one time ST-ZS switching.

Accordingly, two devices among $S_1 \sim S_4$ have two times ACT-ZS switching in each half cycle of i_o . Therefore, the total switching power loss of $S_1 \sim S_4$ caused by ACT-ZS action is expressed as

$$P_{SW,ACT-ZS} = \frac{I_m}{\pi} \times 2f_s \times 2 \times (E_{ON} + E_{OFF} + E_{REC}) \frac{V_{PN}}{V_r I_r}$$

$$= \frac{4I_m}{\pi} (E_{ON} + E_{OFF} + E_{REC}) \frac{V_{PN}f_s}{V_r I_r} \quad (7)$$

TABLE 3. Comparison of switching times in one control cycle for proposed method and traditional SPWM.

			Switching times when $i_o > 0$	Switching times when $i_o < 0$
Proposed method	S_1	ACT-ZS	Two	Zero
		ST-ZS	Two	Two
	S_2	ACT-ZS	Two	Zero
		ST-ZS	Zero	Zero
Traditional SPWM	S_1	ACT-ZS	One	One
		ST-ZS	One	One
	S_2	ACT-ZS	One	One
		ST-ZS	One	One

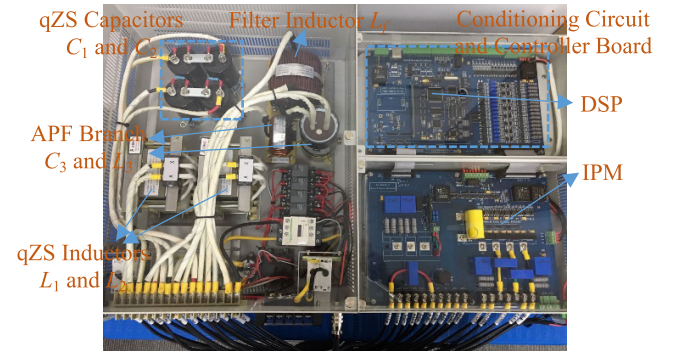


FIGURE 5. Experimental prototype of APF integrated single-phase qZSI.

The power loss caused by ST-ZS behavior can be summarized by two times ST-ZS actions, for two devices among $S_1 \sim S_4$. That is

$$P_{SW,ST-ZS} = 2 \times 2I_L \times (E_{ON} + E_{OFF}) \frac{V_{PN}f_s}{V_r I_r} \quad (8)$$

where E_{ON} and E_{OFF} denote the datasheet values of turn-on and turn-off energy per pulse at reference voltage V_r and current I_r ; E_{REC} is the reverse recovery loss of the anti-parallel diode.

By comparison of (7) and (8) with the ACT and ST switching power loss of traditional SPWM in [33], it can be obtained that the proposed method has the same switching power loss as the traditional SPWM.

The proposed method revises the state from ZS₁ to ZS₂, without other changes of duty cycles. Also, the duty cycles of ACT and ST states maintain as those in the traditional SPWM. Therefore, the computation of conduction power loss of proposed method is the same as that of traditional one [33].

IV. SIMULATION AND EXPERIMENTAL INVESTIGATIONS

The proposed CMV reduction method for the APF integrated single-phase qZSI is simulated in MATLAB/ Simulink, and tested in the built prototype, as shown in Fig. 5. The TMS320F28335 digital signal processor (DSP) based controller board performs the control method; the Intelligent Power Module (IPM) PM50RLA120 is utilized to implement load power phase legs 1 and 2 and the APF phase leg 3. Table 4 lists the system specifications. The PV panel voltage

TABLE 4. System specifications.

Parameter	Value
Rated power of single-phase qZSI, P_m	3 kW
PV panel voltage range, v_{PV}	200~400 V
Grid voltage amplitude, V_g	311 V
Grid voltage frequency, f_0	50 Hz
Switching frequency, f_s	10 kHz
qZS Inductance, L_1 and L_2	500 μ H
qZS capacitance, C_1 and C_2	400 μ F
APF inductance, L_3	500 μ H
APF capacitance, C_3	220 μ F
Filter inductance, L_f	3 mH

range is 200~400 V, implemented by the IT6514C PV simulator while keeping at maximum power point condition. Unity power factor injection into the grid is considered. The qZS inductance and capacitance are 500 μ H and 400 μ F, respectively, at the 10 kHz switching frequency (100 μ s sampling time). The APF capacitance is 220 μ F and filter inductance is 3 mH. Simulation and experimental results using the traditional SPWM and proposed method are compared for the single-phase qZSI, as results shown in Figs. 6-9. In addition, the power losses of qZSI under the two methods are investigated for the SiC-based devices at the 30 kHz switching frequency, and traditional Si-based devices at the 10 kHz switching frequency, respectively, with results shown in Figs. 10 and 11.

A. SIMULATION AND EXPERIMENTAL RESULTS

The PV panel voltage is at 300 V, hence a 0.125 ST duty cycle is performed to get a 400 V dc-link peak voltage, and a 0.78 modulation index is applied to match the 311 V grid voltage amplitude. Fig. 6 (a) shows simulation results of the input current i_L , qZS capacitor- C_1 voltage v_{C1} , and dc-link voltage v_{PN} ; Fig. 6 (b) shows the output current i_o , APF capacitor- C_3 voltage u_C and current i_C ; and Fig. 6 (c) shows the common mode voltage u_{CM} and leakage current i_{LK} of the APF integrated single-phase qZSI with the traditional SPWM. Fig. 7 (a) shows experimental results of the input current i_L , qZS capacitor- C_1 voltage v_{C1} , APF capacitor voltage u_C , and APF branch current i_C ; Fig. 7 (b) shows common mode voltage u_{CM} , qZSI H-bridge output voltage u_H , and output current i_o in 10 ms/div, and that of zoomed view in 100 μ s/div is shown in Fig. 7 (c); Fig. 7 (d) shows harmonic spectrum of qZSI output voltage. Those of proposed method are shown in Figs. 8 and 9, respectively.

From (1), the average input current i_L and qZS capacitor voltage v_{C1} should be 350 V and 7.8 A, respectively. As simulation and experimental results of the proposed method and traditional SPWM shown in Figs. 6 (a), 7 (a), 8 (a) and 9 (a), the average i_L and v_{C1} match the calculations, under the both methods.

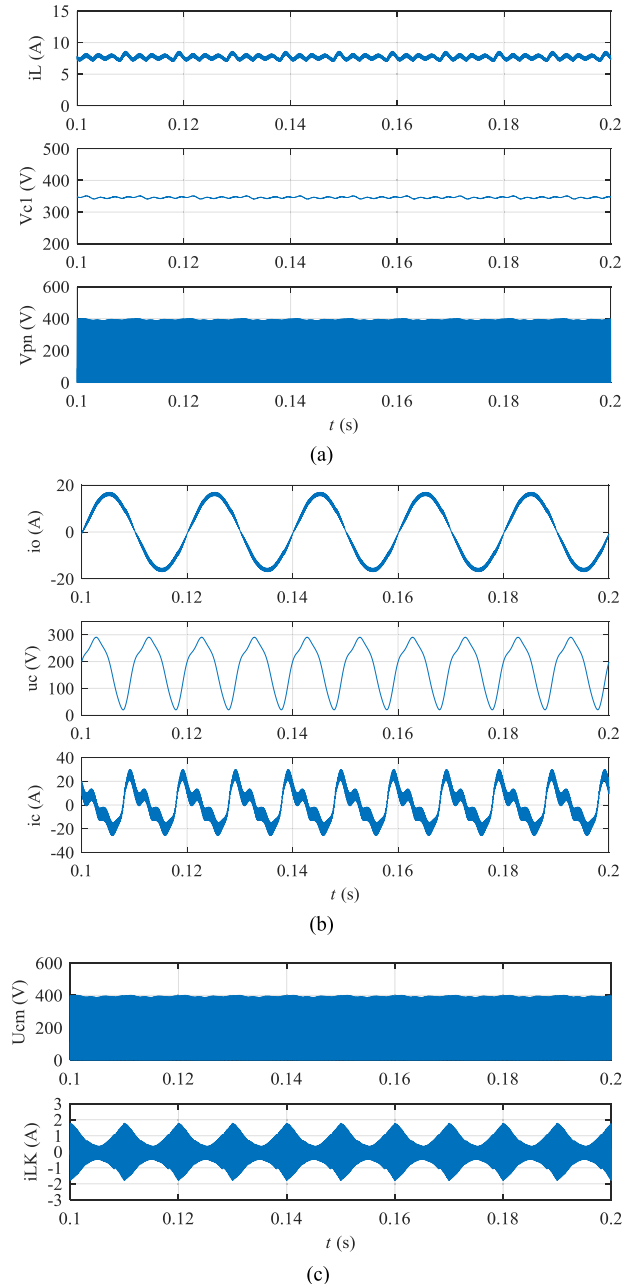


FIGURE 6. Simulation results of APF integrated single-phase qZSI under the traditional SPWM. (a) Input current, qZS capacitor- C_1 voltage, and qZSI dc-link voltage; (b) output current, APF capacitor voltage and current; and (c) common mode voltage and leakage current.

From the input current and qZS capacitor- C_1 voltage in Figs. 6 (a), 7 (a), 8 (a), and 9 (a), as well as APF capacitor voltage and current in Figs. 6 (b), 7 (a), 8 (b), and 9 (a), it can be seen that the dc-side 2ω ripple power are transferred to the APF compensation capacitor in both the proposed method and traditional SPWM. The 2ω ripple ratios of input current, capacitor- C_1 voltage, and dc-link peak voltage are within the limits. The capacitor- C_3 voltage u_C and current i_C fluctuates at the 2ω angle frequency, and the u_C is a dc value with a 2ω component, which matches analysis of (3).

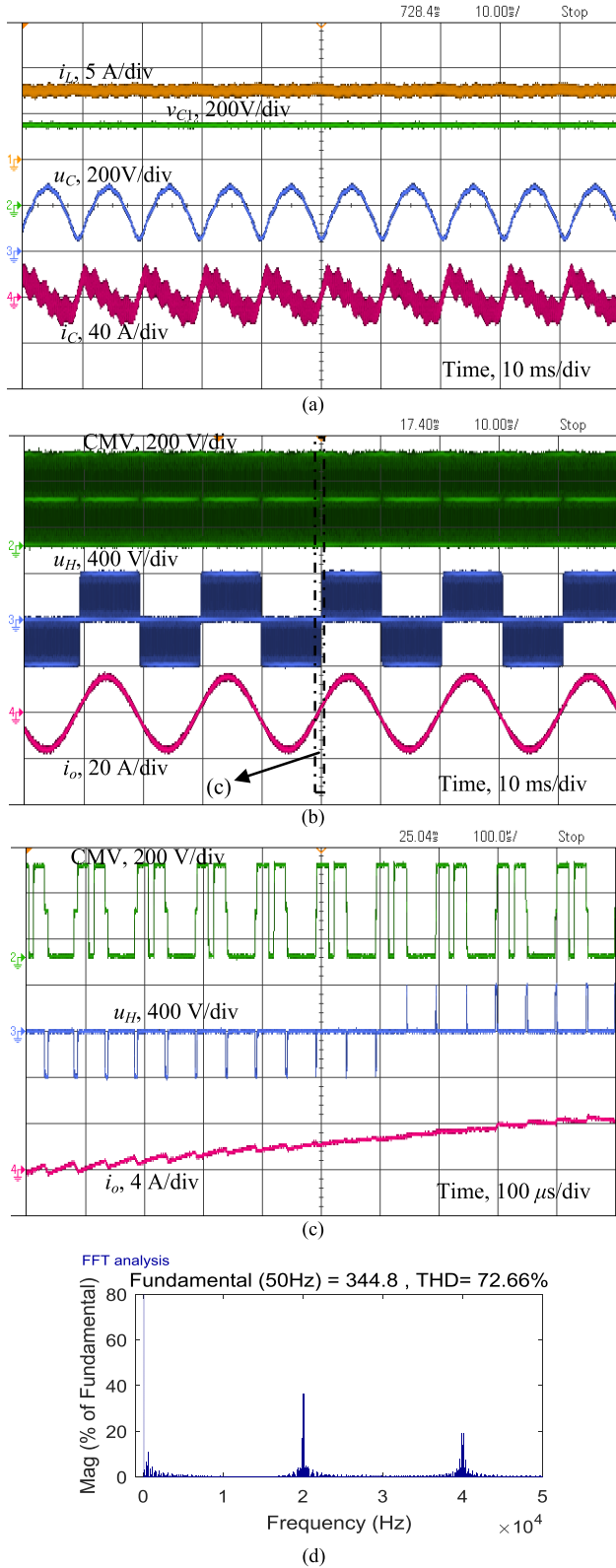


FIGURE 7. Experimental results when the APF integrated single-phase qZSI uses the traditional SPWM. (a) Input current, qZS capacitor- C_1 voltage, APF capacitor voltage, and APF branch current; common mode voltage, qZSI H-bridge output voltage, and output current (b) in 10 ms/div and (c) zoomed in 100 μs/div; (d) harmonic spectrum of qZSI output voltage.

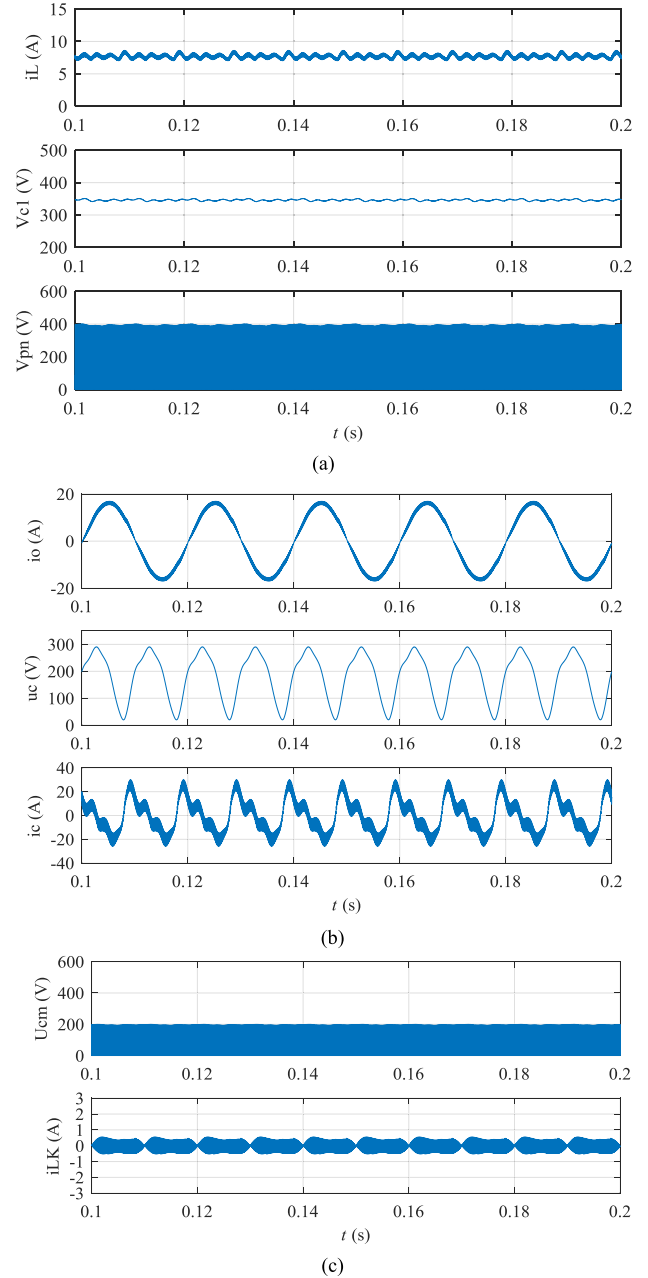


FIGURE 8. Simulation results of APF integrated single-phase qZSI under the proposed method. (a) Input current, qZS capacitor- C_1 voltage, and qZSI dc-link voltage; (b) output current, APF capacitor voltage and current; and (c) common mode voltage and leakage current.

Comparing Figs. 6 (c) and 7 (b) with 8 (c) and 9 (b), it can be seen that the CMV amplitude of the proposed method is reduced half, as a result of a nearly one-fourth reduction of the leakage current amplitude, compared to that of the traditional SPWM. In addition, the CMV behavior of proposed method in Fig. 9 (c) matches the Fig. 4, and that of SPWM matches the Fig. 3, verifying the validity of the analysis and proposed method.

Besides, Figs. 7 (d) and 9 (d) show qZSI output voltage spectrums of the qZSI under the two methods. From the

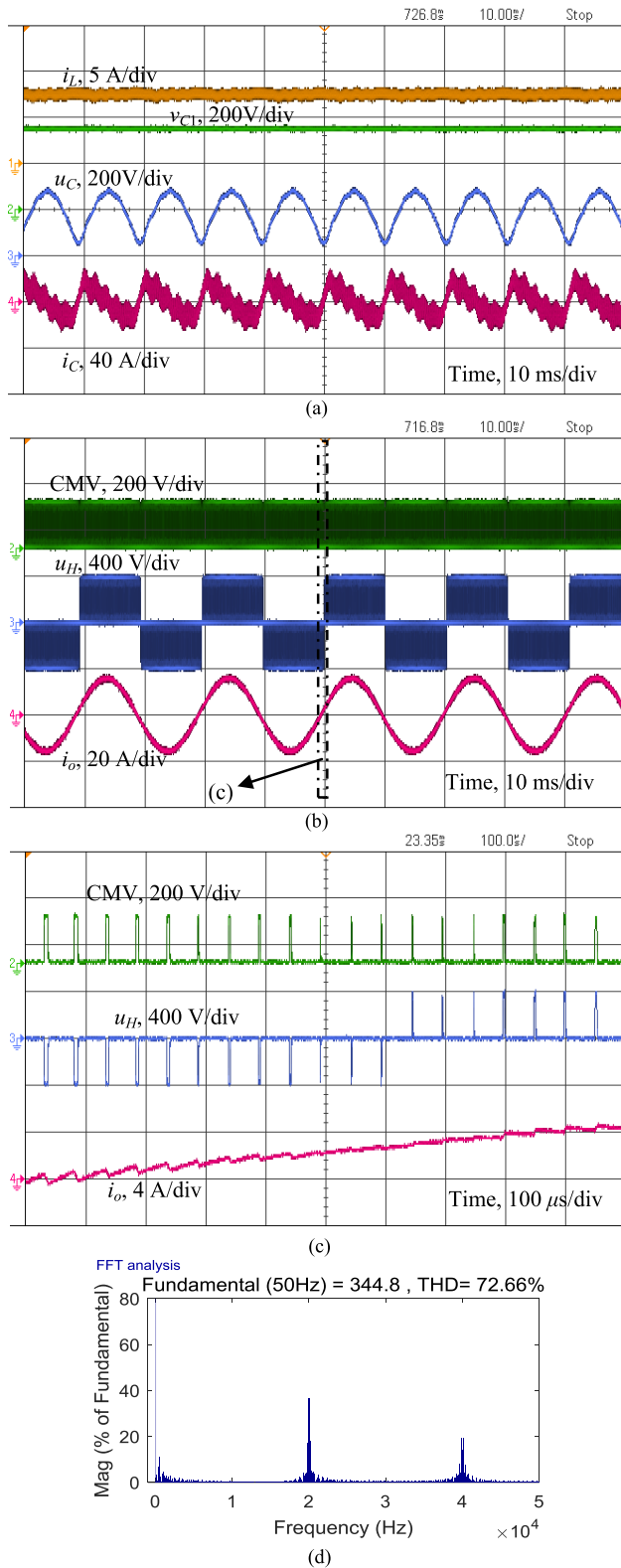


FIGURE 9. Experimental results when the APF integrated single-phase qZSI uses the proposed method. (a) Input current, qZS capacitor- C_1 voltage, APF capacitor voltage, and APF branch current; common mode voltage, qZSI H-bridge output voltage, and APF branch current (b) in 10 ms/div and (c) zoomed in 100 μ s/div; (d) harmonic spectrum of qZSI output voltage.

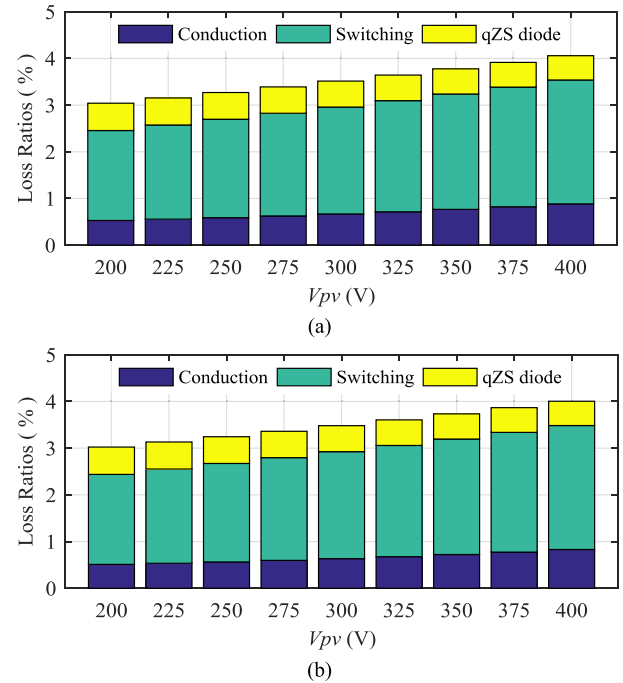


FIGURE 10. Power loss evaluation of Si-based power module for the APF integrated single-phase qZSI. (a) Loss ratios under the traditional SPWM, and (b) loss ratios under the proposed method.

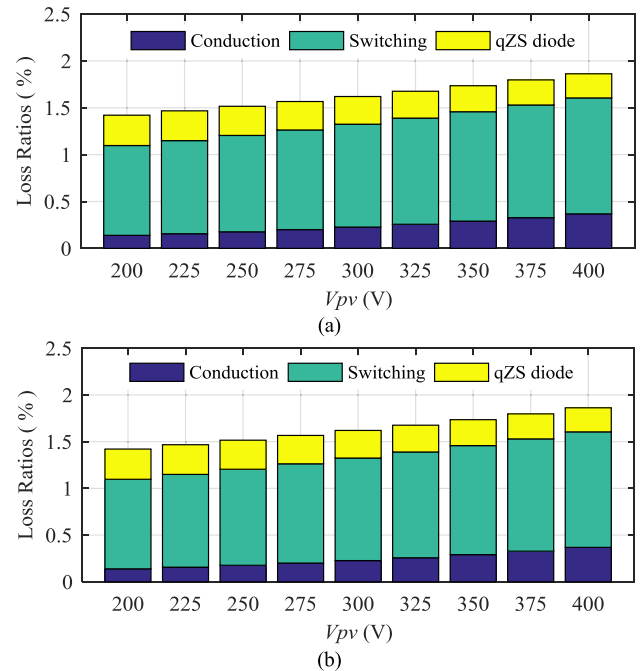


FIGURE 11. Power loss evaluation of SiC-based power module for the APF integrated single-phase qZSI. (a) Loss ratios under the traditional SPWM, and (b) loss ratios under the proposed method.

total harmonic distortion (THD) and amplitudes of harmonic components, it can be seen that the proposed method has the similar voltage spectrums as the SPWM, owing to that the proposed method modifies one of the two zero states

without affecting the active state behaviors, which can also be seen from the zoomed view of output voltage and current in Figs. 7 (c) and 9 (c). The output current has the same increasing and decreasing actions in the two methods. Hence, a same grid filter is applicable for the two methods. In summary, the proposed method decreases the CMV of APF integrated single-phase qZSI without loss of high performance.

B. POWER LOSS ANALYSIS

Based on the system specifications in Table 4, the power loss analysis is performed on the commercialized 1200 V/50 A, full-SiC MOSFET Module CCS050M12CM2 and the Si IPM PM50RLA120 of the prototype. The 30 kHz switching frequency is applied to the former, and that of the later is 10 kHz. The switching power loss and conduction power loss are investigated for the proposed method and traditional SPWM, respectively, under the two modules. Figs. 10 and 11 show loss ratios versus the PV panel voltage variations. The ST duty cycle varies with the PV panel voltage change, to maintain a constant 400 V dc-link peak voltage.

Comparing Fig. 10 (a) with (b), and Fig. 11 (a) with (b), it can be seen that the switching and conduction power losses of the proposed method are same to those of the traditional SPWM, demonstrating the proposed method reducing the CMV without loss of efficiency.

Furthermore, comparing Figs. 10 and 11, it can be seen that by using the SiC power module, the total power loss is less than half compared to using the traditional Si power module. There will also be one-third reduction of passive component values, so less power loss and smaller volume, owing to the 30-kHz switching frequency at the SiC module.

V. CONCLUSION

In this paper, a modified modulation method was proposed for the single-phase APF integrated qZSI PV system to reduce the CMV. By analyzing the switching operation and CMV values of the traditional SPWM, the proposed method revised the zero state ZS_1 into ZS_2 , as a result of half reduction of CMV amplitude and nearly three quarters decrease of leakage current. The switching and conduction power losses of the proposed method were detailed, which was same to those of the traditional SPWM. Simulation and experimental results and power loss computation under SiC and Si power modules demonstrated the CMV and leakage current reduction from the proposed method, while maintaining high performance and efficiency of the APF integrated single-phase qZSI system.

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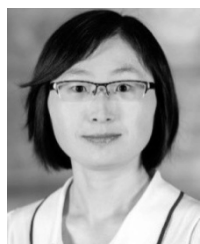
BAOMING GE (M'11) received the Ph.D. degree in electrical engineering from Zhejiang University, Hangzhou, China, in 2000.

He was with the Department of Electrical Engineering, Tsinghua University, Beijing, China, from 2000 to 2002. In 2002, he joined the School of Electrical Engineering, Beijing Jiaotong University, Beijing, where he was promoted to a Professor, in 2006. He was with the University of Coimbra, Coimbra, Portugal, from 2004 to 2005, and with Michigan State University, East Lansing, MI, USA, from 2007 to 2008 and from 2010 to 2014. Then, he joined the Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA. He has authored more than 200 journals and conference papers, two books, and two book chapters, and holds seven patents. His main research interests include renewable energy generation, electrical machine drives, and power electronics.



XIAO LI (S'14–M'19) received the B.Sc. degree in automation from the Harbin Institute of Technology (HIT), China, in 2012, and the Ph.D. degree in electrical engineering from the Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA.

His research interests include power electronics system integration, distributed generations, general power electronic circuit topologies, modeling and control, and wide bandgap devices.

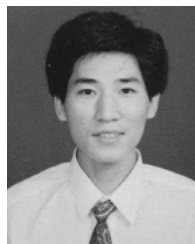


YUSHAN LIU (S'12–M'15–SM'19) received the B.Sc. degree in automation from the Beijing Institute of Technology, Beijing, China, in 2008, and the Ph.D. degree in electrical engineering from the School of Electrical Engineering, Beijing Jiaotong University, Beijing, in 2014.

She was a Postdoctoral Fellow and an Assistant Research Scientist with the Department of Electrical and Computer Engineering, Texas A&M University at Qatar, Doha, Qatar, from 2014 to

2017. She is currently an Associate Professor with the School of Automation Science and Electrical Engineering, Beihang University, Beijing. She has published more than 70 journals and conference papers, one book, and one book chapter in the area of expertise. Her research interests include impedance source inverters, cascade multilevel converters, photovoltaic power integration, renewable energy systems, model predictive control, and smart transformers.

Dr. Liu received the Research Fellow Excellence Award from Texas A&M University at Qatar and the Excellent Doctoral Dissertations Prize from Beijing Jiaotong University. She is an Associate Editor of the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and the IEEE OPEN JOURNAL OF THE INDUSTRIAL ELECTRONICS SOCIETY.



YAOSUO XUE (M'03–SM'12) received the B.Sc. degree in electrical engineering from East China Jiaotong University, Nanchang, China, in 1991, and the M.Sc. degree in electrical engineering from the University of New Brunswick, Fredericton, Canada, in 2004.

From 1991 to 2000, he was an Electrical Engineer-in-Charge with China Railway Design Corporation and led the traction power systems Research and Development of the first China's high-speed railway. From 2005 to 2006, he was with Capstone Turbine Corporation, as a Lead Power Electronics and Systems Engineer. From 2006 to 2008, he was a NSERC PGS-Doctor Scholar with the University of New Brunswick. He was with Siemens Corporate Research, from 2009 to 2015, and established Corporate Technology USA Power Electronics Program. He is currently with the Oak Ridge National Laboratory. His research interests include multilevel converters and smart inverter controls for renewable energy and utility applications.

He is currently an Associate Editor of the IEEE TRANSACTIONS ON POWER ELECTRONICS and the IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS.

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